Cambridge (CIE) A Level Maths: Pure 3

Your notes

Polynomials

Contents

- Polynomial Division
- Factor & Remainder Theorem
- Polynomial Factorisation
- Rational Expressions
- Improper Algebraic Fractions

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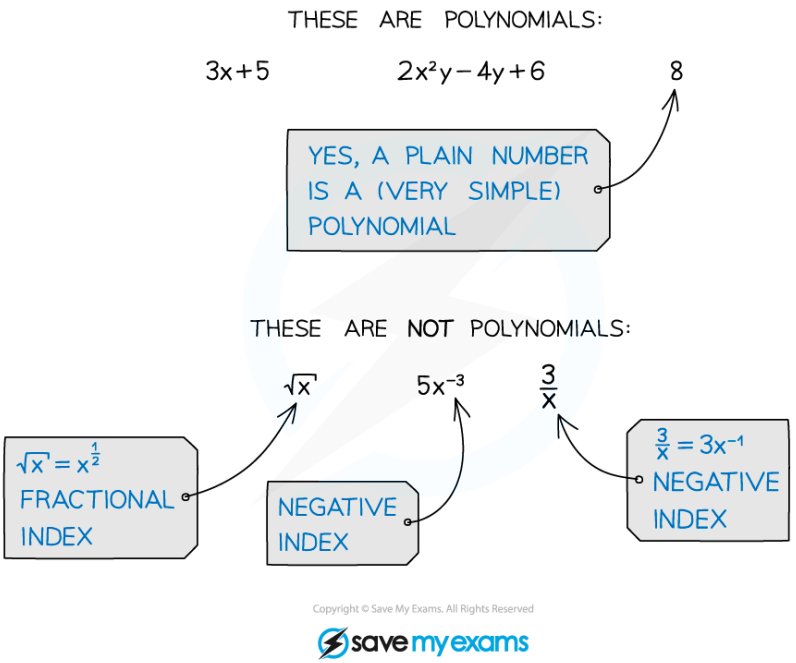
Your notes

Polynomial division

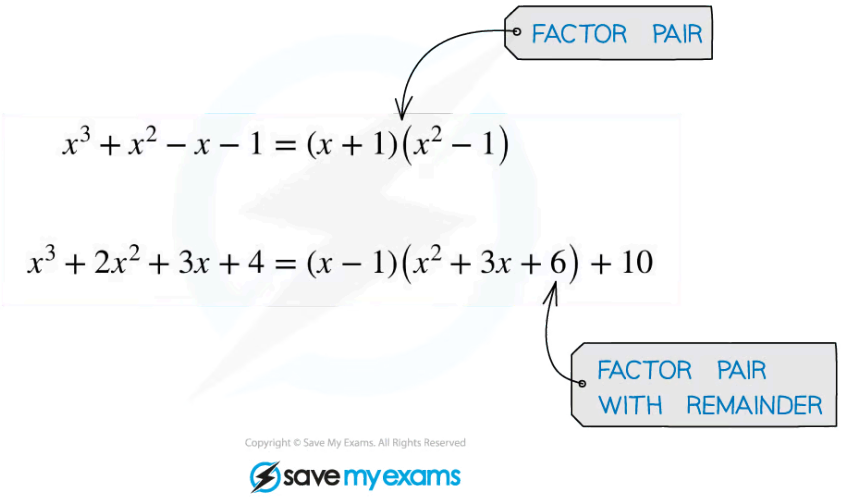
What is a polynomial?

- A **polynomial** is an algebraic expression consisting of a finite number of terms, with non-negative integer indices only



What is polynomial division?

- Polynomial division is a method for splitting polynomials into factor pairs (with or without an accompanying remainder term)



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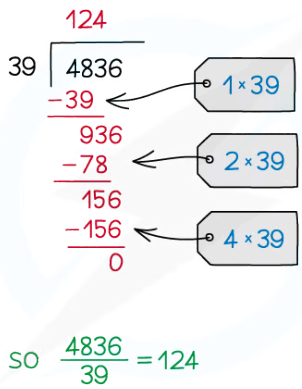
- At A level you will most frequently use it to factorise polynomials, or when dealing with improper (ie 'top-heavy') algebraic fractions

How do I divide polynomials?

- The method used for polynomial division is just like the long division method (sometimes called 'bus stop division') used to divide regular numbers:



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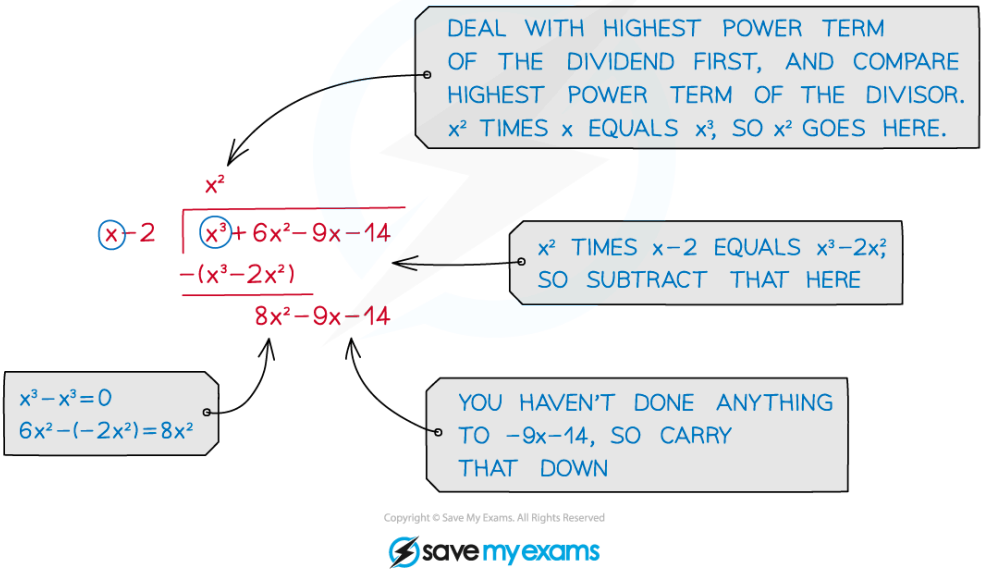
- At A level you will normally be dividing a polynomial dividend of degree 3 or 4 by a divisor in the form **($x \pm p$)**
- The answer to a polynomial division question is built up term by term, working downwards in powers of the variable (usually ***x***)
- Start by dividing by the highest power term
- Write out this multiplied by the divisor and subtract

Your notes

Divide the polynomial $f(x) = x^3 + 6x^2 - 9x - 14$ by $(x - 2)$.



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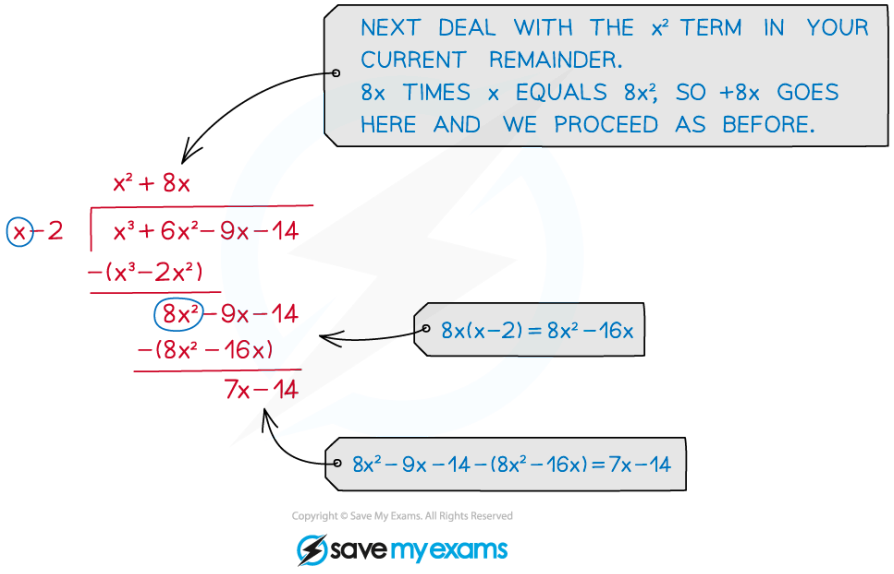
- Continue to divide by each reducing power term and subtracting your answer each time



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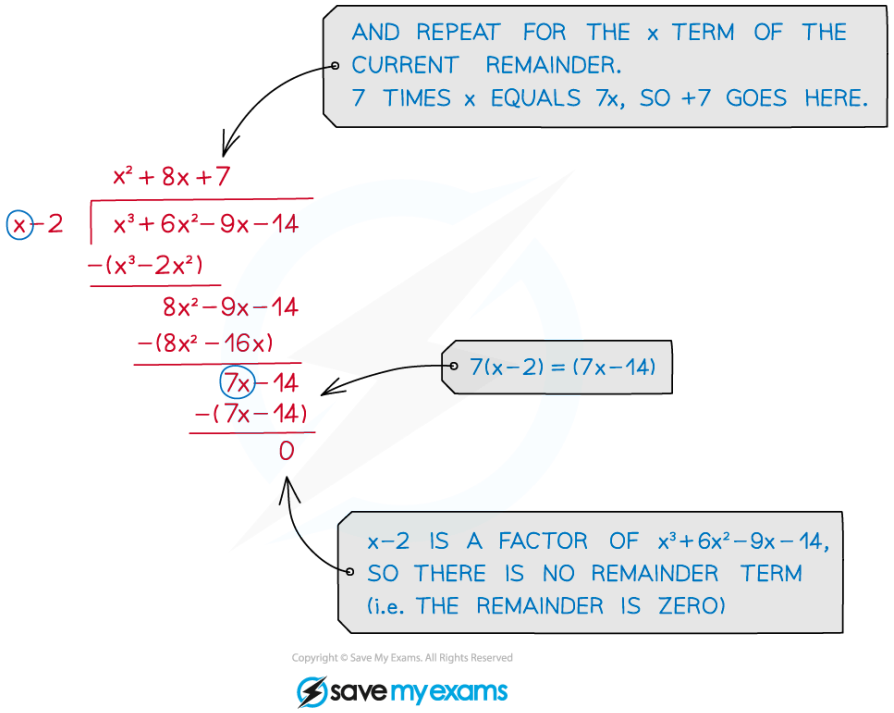
- Continue until you are left with zero



- If the divisor is not a factor of the polynomial then there will be a remainder term left at the end of the division

Worked Example

Your notes

For the polynomial $f(x) = x^3 + 6x^2 - 9x - 14$:

a) Divide $f(x)$ by $(x + 7)$.

b) Find the remainder when $f(x)$ is divided by $(x + 3)$.

a)

$$
\begin{array}{r|l}
 & x^2 - x - 2 \\
\hline
x+7 & x^3 + 6x^2 - 9x - 14 \\
 & -(x^3 + 7x^2) \\
\hline
 & -x^2 - 9x - 14 \\
 & -(-x^2 - 7x) \\
\hline
 & -2x - 14 \\
 & -(-2x - 14) \\
\hline
 & 0
\end{array}
$$

THE x^2 TERM OF THE REMAINDER HERE IS NEGATIVE, SO **$-x$** GOES ON TOP

THE x TERM OF THE REMAINDER HERE IS NEGATIVE, SO **-2** GOES ON TOP

$$\frac{x^3 + 6x^2 - 9x - 14}{x+7} = x^2 - x - 2$$

THIS MEANS THAT $x^3 + 6x^2 - 9x - 14 = (x+7)(x^2 - x - 2)$

b)

$$
\begin{array}{r|l}
 & x^2 + 3x - 18 \\
\hline
x+3 & x^3 + 6x^2 - 9x - 14 \\
 & -(x^3 + 3x^2) \\
\hline
 & 3x^2 - 9x - 14 \\
 & -(3x^2 + 9x) \\
\hline
 & -18x - 14 \\
 & -(-18x - 54) \\
\hline
 & 40
\end{array}
$$

THE x TERM OF THE REMAINDER HERE IS NEGATIVE, SO **-18** GOES ON TOP

THE REMAINDER IS 40

THIS MEANS THAT $x^3 + 6x^2 - 9x - 14 = (x+3)(x^2 + 3x - 18) + 40$

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Quadratic divisor

What is a quadratic divisor?

- Suppose a polynomial of **degree n** is divided by a **quadratic divisor**

- $\dfrac{f(x)}{ax^2+bx+c} = q(x) + \dfrac{r(x)}{ax^2+bx+c}$
- The **quotient** *q* will have degree $n-2$
- The degree of the **remainder** *r* will be less than 2
  - It could be degree 1 (linear)
  - Or it could be degree 0 (constant)

How do I divide a polynomial by a quadratic divisor?

- You use **polynomial division!**
- **Step 1**
  Divide the **leading term** of the **polynomial** by the **squared term** of the **divisor**
  - This gives the **leading term** of the **quotient**
- **Step 2**
  Multiply this term by the **divisor**
- **Step 3**
  Subtract this from the polynomial to get a **new polynomial** with a **lower degree**
- **Continue** these steps until you have an expression with a degree lower than 2

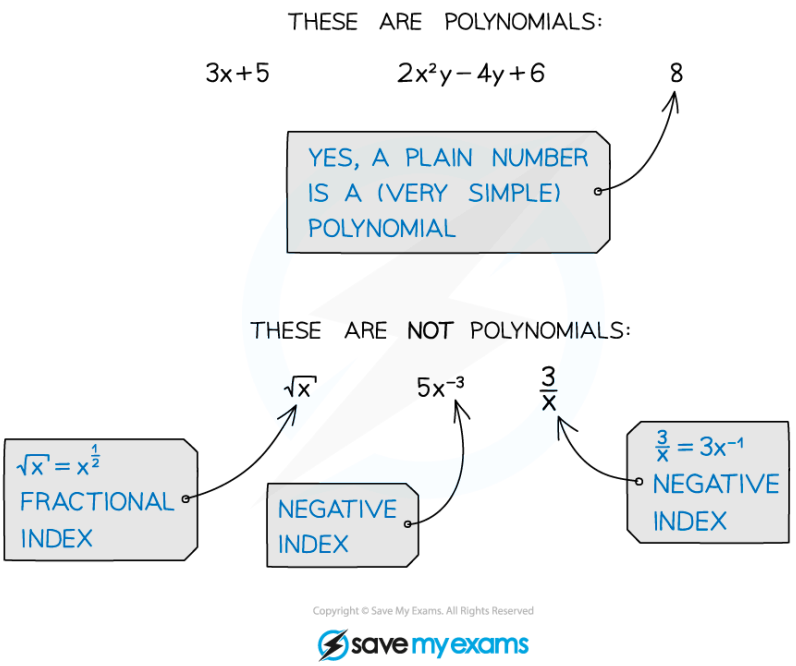
Your notes

Your notes

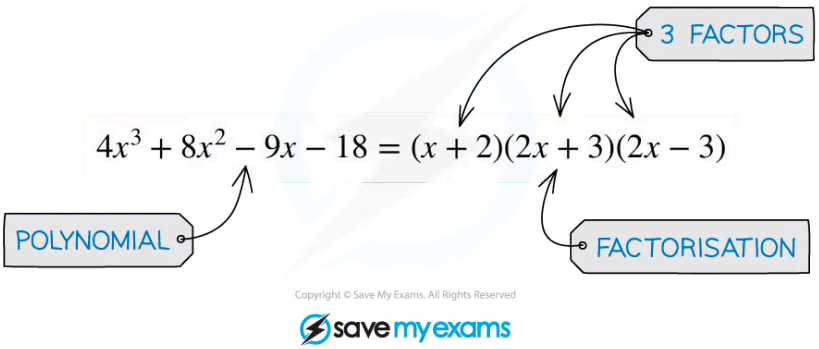
Factor theorem

What is the factor theorem?

- The factor theorem is a very useful result about polynomials
- A **polynomial** is an algebraic expression consisting of a finite number of terms, with non-negative integer indices only



- At A level you will most frequently use the factor theorem as a way to simplify the process of factorising polynomials



How do I apply the factor theorem?

- For a polynomial **f(x)** the factor theorem states that:
  - If **f(p) = 0**, then **(x – p)** is a factor of **f(x)**

  AND

  - If **(x – p)** is a factor of **f(x)**, then **f(p) = 0**

$$g(x) = 2x^3 - 7x^2 + 9$$
$$g(3) = 2(3)^3 - 7(3)^2 + 9$$
$$= 54 - 63 + 9$$
$$= 0$$

g(3) = 0

BY (i) THE FACTOR THEOREM TELLS US THAT (x−3) IS A FACTOR OF g(x)

$$h(x) = 4x^3 + 8x^2 - 9x - 18$$
$$= (x+2)(4x^2 - 9)$$

(x+2) = (x−(−2)) IS A FACTOR

BY (ii) THE FACTOR THEOREM TELLS US THAT h(−2) = 0 (TEST IT OUT!)

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Your notes

Examiner Tips and Tricks

- In an exam, the values of ***p*** you need to find that make **f(p) = 0** are going to be integers close to zero.
  - Try **p** = 1 and -1 first, then 2 and -2, then 3 and -3
  - It is very unlikely that you'll have to go beyond that.

Worked Example

Your notes

For the polynomial $f(x) = x^3 + 6x^2 - 9x - 14$:

a) Use the factor theorem to show that $(x - 2)$ is a factor of $f(x)$.

b) Use the factor theorem to find another factor of $f(x)$.

a) $f(2) = (2)^3 + 6(2)^2 - 9(2) - 14$
$= 8 + 24 - 18 - 14 = 0$

FOR $(x-2)$ TO BE A FACTOR, $f(2)$ MUST BE EQUAL TO ZERO, NOT $f(-2)$

SO $(x-2)$ IS A FACTOR OF $f(x)$

b) $f(1) = (1)^3 + 6(1)^2 - 9(1) - 14$
$= 1 + 6 - 9 - 14 = -16$

TRY $x=1$ FIRST

$f(-1) = (-1)^3 + 6(-1)^2 - 9(-1) - 14$
$= -1 + 6 + 9 - 14 = 0$

$f(1) \neq 0$, SO TRY $x=-1$ NEXT

SO $(x+1)$ IS ANOTHER FACTOR OF $f(x)$

$f(-1) = 0$, SO BY THE FACTOR THEOREM $(x-(-1)) = (x+1)$ IS A FACTOR OF $f(x)$

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Remainder theorem

What is the remainder theorem?

- The **factor theorem** is actually a special case of the more general **remainder theorem**
- The **remainder theorem** states that when the polynomial f(x) is divided by (x - a) the remainder is f(a)
  - You may see this written formally as f(x) = (x - a)Q(x) + f(a)
  - In **polynomial division**
    - Q(x) would be the **result** (at the top) of the division (the **quotient**)
    - f(a) would be the **remainder** (at the bottom)
    - (x - a) is called the **divisor**
  - In the case when f(a) = 0, f(x) = (x - a)Q(x) and hence (x - a) is a factor of f(x)– the **factor theorem**!

How do I apply the remainder theorem?

Your notes

- If it is the **remainder** that is of particular interest, the remainder theorem saves the need to carry out polynomial division in full
  - e.g. The remainder from $(x^2 - 2x) \div (x - 3)$ is $3^2 - 2 \times 3 = 3$
  - This is because if f(x) = $x^2 - 2x$ and a = 3
- If the remainder from a polynomial division is known, the **remainder theorem** can be used to find unknown coefficients in polynomials
  - g. The remainder from $(x^2 + px) \div (x - 2)$ is 8 so the value of *p* can be found by solving $2^2 + p(2) = 8$, leading to $p = 2$
  - In harder problems there may be more than one unknown in which case simultaneous equations would need setting up and solving
- The more general version of remainder theorem is if f(x) is divided by (ax - b) then the remainder is $\mathrm{f}\left(\frac{b}{a}\right)$
  - The shortcut is still to evaluate the polynomial at the value of x that makes the divisor (ax - b) zero but it is not necessarily an integer

Worked Example

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Your notes

The polynomial $p(x)$ is given by $8x^4 + ax^2 + bx - 1$,
where a and b are integer constants.
When $p(x)$ is divided by $(x - 1)$ the remainder is 9.
When $p(x)$ is divided by $(2x - 1)$ the remainder is 1.
Find the values of a and b.

REMAINDER THEOREM: "f(a) IS THE REMAINDER WHEN f(x) IS DIVIDED BY (x − a)"

$x - 1 = 0$ WHEN $x = 1$

$\therefore p(1) = 9$

$8 + a + b - 1 = 9$

$a + b = 2$

AND

$2x - 1 = 0$ WHEN $x = \frac{1}{2}$

$p\left(\frac{1}{2}\right) = 1$

$8\left(\frac{1}{2}\right)^4 + a\left(\frac{1}{2}\right)^2 + b\left(\frac{1}{2}\right) - 1 = 1$

$\frac{1}{2} + \frac{1}{4}a + \frac{1}{2}b - 1 = 1$

$a + 2b = 6$

SOLVING SIMULTANEOUSLY:

$a + b = 2$

$a + 2b = 6$

$b = 4$

$a = -2$

$a = -2, \ b = 4$

$p(x) = 8x^4 - 2x^2 + 4x - 1$

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Examiner Tips and Tricks

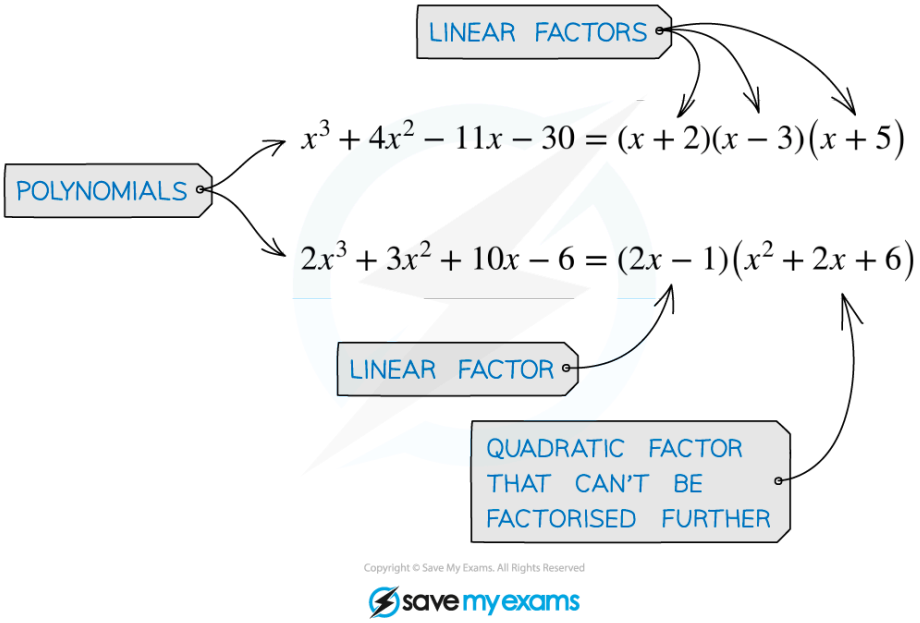
- Exam questions will use formal mathematical language which can make factor and remainder theorem questions sound more complicated than they are.
- Ensure you are familiar with the various terms from these revision notes

Your notes

Polynomial factorisation

What is polynomial factorisation?

- Factorising a polynomial combines the **factor theorem** with the method of **polynomial division**
- The goal is to break down a polynomial as far as possible into a product of linear factors



How do I factorise a polynomial?

- At A level you will usually be asked to factorise a **cubic** – i.e. a polynomial where the highest power of ***x*** is 3

THESE ARE ALL EXAMPLES OF CUBICS:

$$x^3 + 2x^2 - 5x + 7$$

$$3 - x^2 - 4x^3$$

$$x^3 + 2x$$

$$2x^3$$

- To factorise a cubic polynomial **f(x)** follow the following steps:

Your notes

> Fully factorise $x^3 + 4x^2 - 11x - 30$.

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- Step 1. Find a value ***p*** that makes **f(*p*) =** 0

$(1)^3 + 4(1)^2 - 11(1) - 30 = -36$ ← TRY $x=1$

$(-1)^3 + 4(-1)^2 - 11(-1) - 30 = -16$ ← TRY $x=-1$

$(2)^3 + 4(2)^2 - 11(2) - 30 = -28$ ← TRY $x=2$

$(-2)^3 + 4(-2)^2 - 11(-2) - 30 = 0$ ✓ ← TRY $x=-2$

BY THE FACTOR THEOREM THIS TELLS US THAT $(x-(-2)) = (x+2)$ IS A FACTOR OF THE POLYNOMIAL

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- Step 2. Use polynomial division to divide **f(x)** by **(*x* – *p*)**

$$
\begin{array}{r|l}
 & x^2+2x-15 \\
\hline
x+2 & x^3+4x^2-11x-30 \\
 & -(x^3+2x^2) \\
\hline
 & 2x^2-11x-30 \\
 & -(2x^2+4x) \\
\hline
 & -15x-30 \\
 & -(-15x-30) \\
\hline
 & 0
\end{array}
$$

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- Step 3. Use the result of your division to write

$$f(x) = (x - p)(ax^2 + bx + c)$$

$$\text{SO} \quad x^3 + 4x^2 - 11x - 30 = (x+2)(x^2 + 2x - 15)$$

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- Step 4. If the quadratic $(ax^2 + bx + c)$ is factorisable, factorise it and write $f(x)$ as a product of three linear factors (if the quadratic is not factorisable, then your result from Step 3 is the final factorisation)

$$x^2 + 2x - 15 = (x+5)(x-3)$$

$$\text{SO} \quad x^3 + 4x^2 - 11x - 30 = (x+2)(x+5)(x-3)$$

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Your notes

Examiner Tips and Tricks

- The method outlined above can be logically extended to factorise a polynomial of any degree.

Worked Example

Your notes

Fully factorise $x^3 + 6x^2 - 9x - 14$.

STEP 1:

$f(1) = (1)^3 + 6(1)^2 - 9(1) - 14$
$= 1 + 6 - 9 - 14 = -16$

TRY $x=1$ FIRST

$f(-1) = (-1)^3 + 6(-1)^2 - 9(-1) - 14$
$= -1 + 6 + 9 - 14 = 0$

$f(1) \neq 0$, SO TRY $x=-1$ NEXT

SO $(x+1)$ IS A FACTOR OF $f(x)$

$f(-1) = 0$, SO BY THE FACTOR THEOREM $(x-(-1)) = (x+1)$ IS A FACTOR OF $f(x)$

STEP 2:

$$
\begin{array}{r|l}
 & x^2 + 5x - 14 \\
\hline
x+1 & x^3 + 6x^2 - 9x - 14 \\
 & -(x^3 + x^2) \\
\hline
 & 5x^2 - 9x - 14 \\
 & -(5x^2 + 5x) \\
\hline
 & -14x - 14 \\
 & -(-14x - 14) \\
\hline
 & 0
\end{array}
$$

STEP 3:

SO $x^3 + 6x^2 - 9x - 14 = (x+1)(x^2 + 5x - 14)$

STEP 4:

$(x^2 + 5x - 14) = (x+7)(x-2)$

SO $x^3 + 6x^2 - 9x - 14 = (x+1)(x+7)(x-2)$

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Your notes

Rational expressions

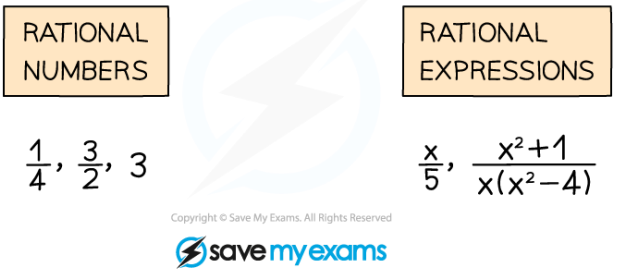
What are rational expressions?

- **Rational** numbers are numbers that can be written as a fraction (**quotient**)

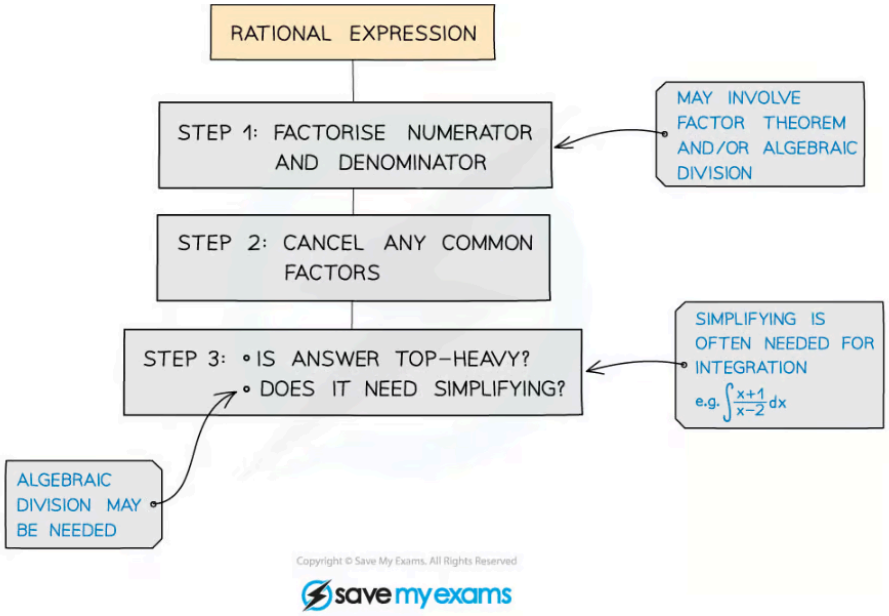
  Rational comes from **ratio** – a number is rational if it can be written as a ratio of two integers – ie a fraction!

- A **rational expression** is an **algebraic fraction**

  The ratio between two algebraic expressions (usually polynomials)



How do I simplify rational expressions?



- You may need to recall the **factor theorem**

Your notes

FACTOR THEOREM

IF $f(a)=0$ THEN $(x-a)$ IS A FACTOR OF $f(x)$

i.e. $f(x) = (x-a)(\ldots\ \ldots)$

EXTENDED FACTOR THEOREM

IF $f\left(\frac{b}{a}\right)=0$ THEN $(ax-b)$ IS A FACTOR OF $f(x)$

i.e. $f(x) = (ax-b)(\ldots\ \ldots)$

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- Start by **factorising** polynomials using **factor theorem** or **algebraic division**

Simplify $\dfrac{x^3-7x+6}{x^2+2x-3}$

$f(x) = x^3-7x+6$

$f(1) = 1^3-7\times1+6=0$

$\therefore (x-1)$ IS A FACTOR OF $f(x)$

FACTOR THEOREM

$$
\begin{array}{r|l}
 & x^2+x-6 \\
\hline
x-1 & x^3+0x^2-7x+6 \\
 & \underline{x^3-x^2} \\
 & \quad x^2-7x \\
 & \quad \underline{x^2-x} \\
 & \qquad -6x+6 \\
 & \qquad \underline{-6x+6} \\
 & \qquad\quad 0
\end{array}
$$

ALGEBRAIC DIVISION

$\therefore f(x) = (x-1)(x^2+x-6)$

$f(x) = (x-1)(x+3)(x-2)$

FACTORISE THE QUADRATIC

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- **cancel** any common (linear) factors

Your notes

$$\frac{x^3-7x+6}{x^2+2x-3} = \frac{\cancel{(x-1)}\cancel{(x+3)}(x-2)}{\cancel{(x+3)}\cancel{(x-1)}}$$

$$= (x-2)$$

FROM EARLIER

CANCEL COMMON FACTORS

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- **recognise** a **top-heavy (improper)** rational expression, simplify if needed

Worked Example

Simplify $\dfrac{8x^3+40x^2+48x}{2x^3+4x^2-6x}$

$$\frac{8x^3+40x^2+48x}{2x^3+4x^2-6x} = \frac{8x(x^2+5x+6)}{2x(x^2+2x-3)}$$

$$= \frac{\overset{4}{\cancel{8}}\cancel{x}\cancel{(x+3)}(x+2)}{\cancel{2}\cancel{x}\cancel{(x+3)}(x-1)}$$

$$= \frac{4(x+2)}{x-1}$$

STEP 1 FACTORISE TOP & BOTTOM

STEP 2 CANCEL COMMON FACTORS

STEP 3 THIS IS TOP HEAVY BUT NO NEED TO SIMPLIFY ANY FURTHER

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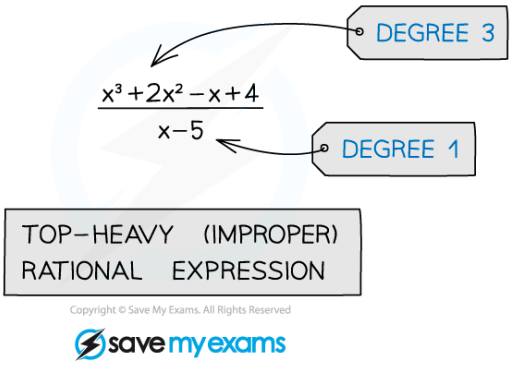
Your notes

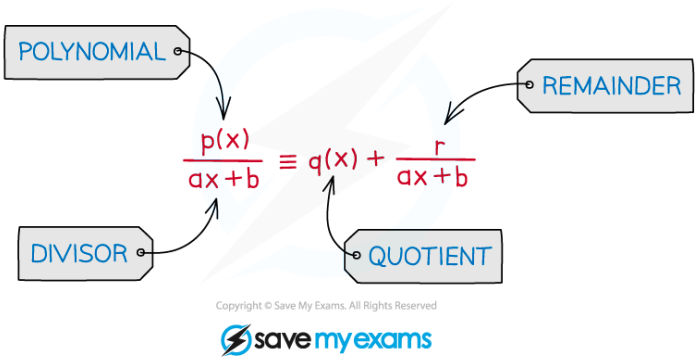
Improper algebraic fractions

What are improper algebraic fractions?

- The **degree** of the numerator is **greater than** or **equal** to the degree of the denominator



How do I simplify improper algebraic fractions?



- Write as a **quotient** and a **remainder**
- The algebraic equivalent of changing a top-heavy fraction to a mixed number

Your notes

$$\frac{x^3+3x^2-4x+2}{x-1}$$

$$\begin{array}{r|l} & x^2+4x+0 \\ \hline x-1 & x^3+3x^2-4x+2 \\ & x^3-x^2 \\ \hline & \quad 4x^2-4x \\ & \quad 4x^2-4x \\ \hline & \qquad 0x+2 \\ & \qquad 0x+0 \\ \hline & \qquad\quad 2 \end{array}$$

STEP 1
DIVIDE TOP BY BOTTOM USING ALGEBRAIC DIVISION

$$\frac{x^3+3x^2-4x+2}{x-1} = x^2+4x+\frac{2}{x-1}$$

STEP 2
WRITE ANSWER AS A QUOTIENT AND A REMAINDER

THIS IS THE ALGEBRAIC EQUIVALENT OF MIXED NUMBERS, e.g. $\frac{17}{5} = 3\frac{2}{5}$
$(17 \div 5 = 3 \text{ r}2)$

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Examiner Tips and Tricks

Remember that simple cases are sometimes the hardest to spot!

$$\frac{x}{x+1}$$

TOP-HEAVY AS NUMERATOR AND DENOMINATOR BOTH DEGREE 1

$$= 1 - \frac{1}{x+1}$$

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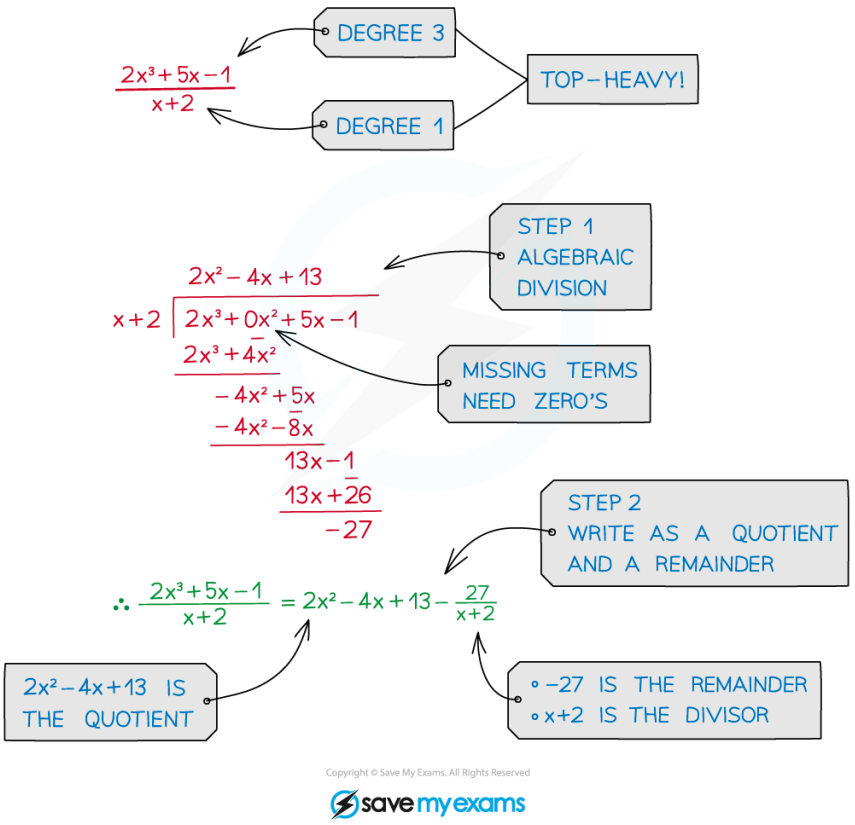
Worked Example

Your notes

Work out $(2x^3 + 5x - 1) \div (x + 2)$